# Virtual Private Networks

IE3122: Network Security | Week 9

*Communicating confidentially over a network built for neither privacy nor trust*

# Today's Agenda

- The nature of the internet, and why confidentiality has to be added, not assumed
- What a VPN is and how it works
- The concept of a tunnel, and how it differs from a VPN
- Software versus hardware VPNs, and the five VPN types, each analysed in depth
- IPsec and SSL VPN in detail, including SSL VPN's modular building blocks
- An overview of how VPNs and tunnels are actually configured on Cisco devices

# Learning Outcomes for Today

**LO3**

- Analyse security requirements and recommend appropriate technologies.

**LO4**

- Evaluate different security technologies, compare them, and identify appropriate use cases.

- *By the end of this session you should be able to explain what makes a VPN a VPN and not just a tunnel, recommend the right VPN type for a given scenario, and describe how IPsec and SSL VPN actually differ under the hood.*

# Foundations

*Why confidentiality has to be built in, not assumed*

# The Nature of the Internet

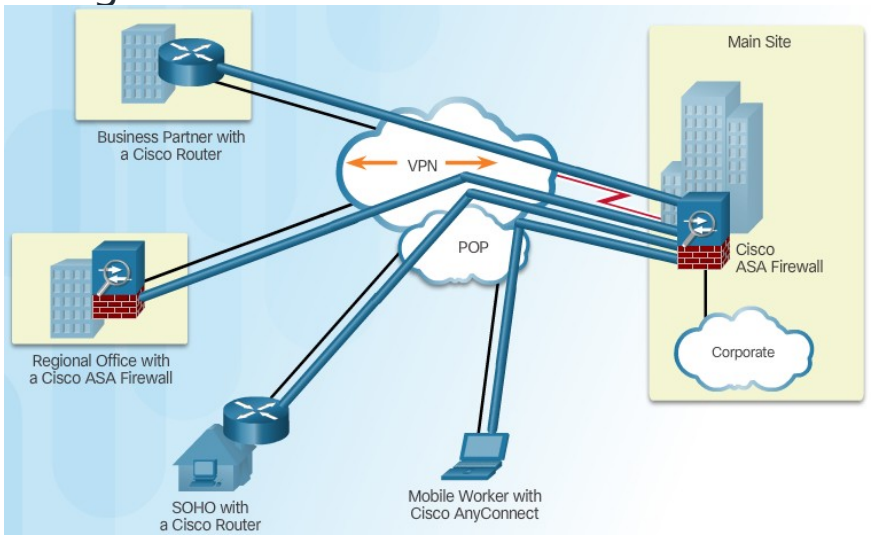
- The internet is a public, packet-switched network of interconnected networks, built from the start to prioritise connectivity and reachability, not privacy.
- A packet leaving your device typically crosses many intermediate hops, home router, ISP, internet exchange, another ISP, before reaching its destination, and you control or trust almost none of them.
- Recall Week 4 and Week 6: ARP spoofing, TCP hijacking, and simple traffic sniffing are all real, demonstrated ways that data crossing an untrusted segment can be intercepted or altered.
- At the same time, organisations increasingly need to connect branch offices, remote employees, and cloud resources across exactly this same untrusted public infrastructure. That tension is the entire reason VPNs exist.

# The Need for Communication Confidentiality

- Confidentiality means only the intended recipient can read the data, even if someone else captures it in transit.
- Without it, financial data, credentials, health records, and ordinary business communication all cross the same public infrastructure in a form anyone positioned along the path could potentially read.
- This isn't a hypothetical risk specific to VPNs. It's the same underlying weakness this whole module keeps returning to: protocols and infrastructure designed for connectivity, not verification or secrecy, by default.

# What Is a VPN?

- A Virtual Private Network (VPN) creates a secure, encrypted connection across a public network, so the connected parties can communicate as if they were on the same private, trusted network.
- It achieves this through three things working together: tunneling (encapsulating traffic), encryption (confidentiality), and authentication (verifying the parties involved).
- The result typically provides confidentiality, integrity, and authentication all at once, properties the underlying public network never guaranteed on its own.



# How a VPN Actually Works

- A VPN client or gateway first establishes a secure session with a VPN server or peer gateway, usually via a key exchange process that authenticates both sides and agrees on cryptographic parameters.
- The original packet is encapsulated inside a new packet, giving it a new outer header suited to crossing the public network.
- That encapsulated packet is then encrypted, so its contents are unreadable to anything observing the public network in between.
- The encrypted, encapsulated packet travels across the public network to the far-end VPN device, which decrypts it, removes the outer encapsulation, and delivers the original packet to its real destination.

# How a VPN Actually Works

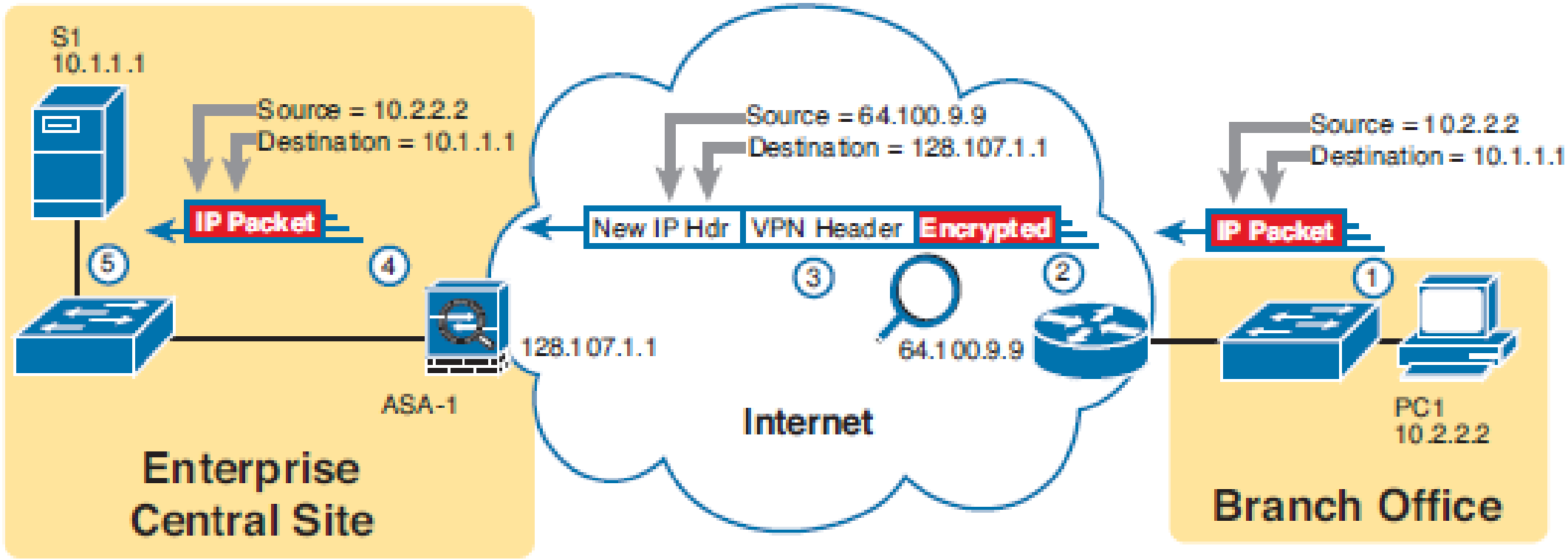



**Figure 7-1** *VPN Tunnel Concepts for a Site-to-Site Intranet VPN*

# The Concept of a Tunnel

- A tunnel is a mechanism for encapsulating one protocol's packets inside another protocol's packets, so they can be carried across a network that might not otherwise support or route them directly.
- A tunnel is purely a transport and encapsulation mechanism. On its own, it does not provide encryption, confidentiality, or authentication of any kind.
- GRE (Generic Routing Encapsulation) is the classic example: it can carry a wide variety of traffic, including protocols IP itself wouldn't normally route, wrapped inside an ordinary IP packet, entirely in plain text.

# Tunnel vs VPN: Not the Same Thing

- A tunnel is just encapsulation. A VPN specifically adds security, encryption, integrity, and authentication, on top of that encapsulation.
- This means not every tunnel is a VPN. A plain GRE tunnel carries traffic across the internet, but that traffic is still sitting in plain text the entire way; anyone intercepting it can read it directly.
- Every VPN does rely on some form of tunneling to carry traffic, but a VPN is defined by the security layered on top of that tunnel, not by the encapsulation alone.
- A very common real-world design combines both deliberately: a GRE tunnel carries traffic (including things plain IPsec alone can't, like dynamic routing protocols or multicast), and IPsec then encrypts that GRE traffic. This is often called GRE over IPsec, and you'll see its configuration later today.

# Tunnel vs VPN: Comparison

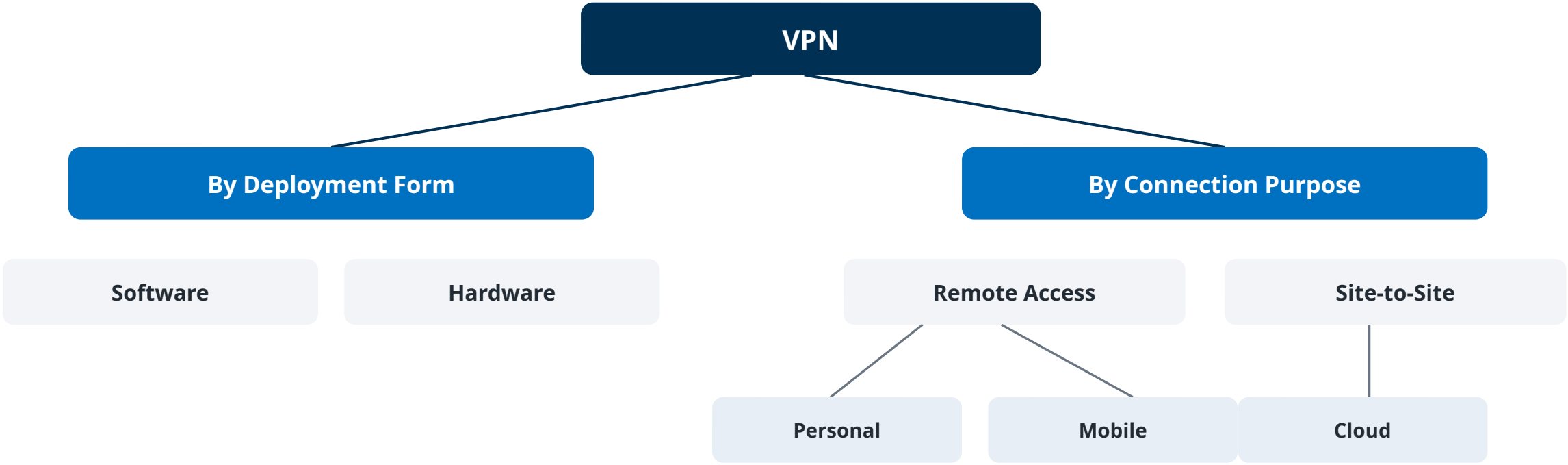
| Aspect | Tunnel | VPN |
|---|---|---|
| Primary purpose | Encapsulate one protocol inside another for transport | Secure, confidential communication over a public network |
| Encryption | Not inherent, unless separately added | Always included |
| Authentication | Not inherent | Typically included |
| Example | GRE, IP-in-IP | IPsec, SSL/TLS VPN, GRE over IPsec |

# How VPNs Are Classified

*Two different questions, not one flat list of types*

CLASSIFICATION

# A Taxonomy, Not a Flat List




*Deployment form and connection purpose are independent questions. Personal and Mobile VPNs are variants of Remote Access; Cloud VPN is mostly a variant of Site-to-Site. Keep this in mind through the next section, since it explains why some of the five types will sound familiar once you've seen the two they build on.*

# Software VPN

- Implemented as a software application or client running on a general-purpose device, or as software-based functionality on a standard server, rather than on dedicated purpose-built hardware.
- Examples: OpenVPN client software, the Cisco AnyConnect client, built-in operating system VPN clients, and software-based VPN server applications running on ordinary servers.
- Flexible and comparatively cheap to deploy widely, but relies on the host device's own processor for encryption, which can become a real performance bottleneck on underpowered devices.

# Hardware VPN

- Implemented via a dedicated physical appliance, a VPN concentrator, or a router or firewall with VPN capability built into dedicated hardware, often including cryptographic acceleration chips.
- Examples: Cisco ISR or ASR routers with VPN support, dedicated VPN concentrators, and ASA firewalls.
- Higher upfront cost, but significantly higher sustained throughput thanks to dedicated crypto hardware, and centralises management at the network edge rather than on every individual device.

# Software vs Hardware VPN: Comparison

| Aspect | Software VPN | Hardware VPN |
|---|---|---|
| Cost | Lower upfront cost | Higher upfront cost |
| Performance | Limited by the host device's own CPU | High, with dedicated crypto acceleration |
| Scalability | Easy to deploy across many individual endpoints | Centralised; fewer devices, each handling more traffic |
| Typical fit | Remote Access, individual users | Site-to-Site, whole-network connections |

# Five VPN Types

*Remote Access and Site-to-Site, and three variants that build on them*

TYPE 1

# Remote Access VPN: Overview

- Connects a single remote user's device to a private organisational network across the internet, so that device behaves as though it were physically on that network.
- The user runs VPN client software, authenticates to a VPN gateway at the organisation's edge, and an encrypted tunnel is established. Depending on configuration, either all of the device's traffic routes through this tunnel, or only traffic destined for the organisation's own network does (split tunneling).
- Typical use case analysis: a mid-sized accounting firm has staff who need to work from home some days, reaching internal file servers and a practice-management application, without the firm buying or managing hardware for every employee's house.
  - Many unpredictable individual locations, not fixed sites, needs a VPN type built around individual client-initiated connections rather than fixed gateway-to-gateway links.
  - Low upfront cost, since there's no dedicated office to equip, needs a mostly software-based client rather than hardware at every location.
  - Staff need full access to internal resources as if physically in the office, needs the tunnel to make the remote device behave as though directly on the internal network.
  - IT wants centralised control over who connects and when, needs authentication to happen at one central gateway, not scattered across many individual devices.

# Remote Access VPN: Analysis

- Hardware/software: primarily software on the client side; the gateway can be software or a dedicated hardware concentrator or firewall.
- Throughput: generally modest per individual user, but the gateway's aggregate capacity across many simultaneous users matters far more than any single connection.
- Limitations: entirely dependent on the remote user's own internet connection quality; client software must be installed and kept updated on every device; a compromised remote endpoint extends real trust into the network from a device the organisation doesn't fully control.
- Cost: relatively low per user, scaling mainly with the number of concurrent connections the gateway must support.
- Typical use cases: employees working from home or travelling, needing secure access to internal company resources.

# Remote Access VPN: Summary

| Dimension | Remote Access VPN |
|---|---|
| Hardware/Software | Software client; gateway is software or hardware |
| Throughput | Modest per user; aggregate capacity matters at scale |
| Limitations | Depends on user's connection; extends trust to unmanaged devices |
| Cost | Low per user |
| Typical use case | Remote and travelling employees |

TYPE 2

# Site-to-Site VPN: Overview

- Connects two entire networks, such as two branch offices, or a branch and a data centre, permanently across the internet, so devices on either side can reach each other as though on one network.
- VPN gateways at each site establish a persistent encrypted tunnel between them. No client software is needed on individual end-user devices; the gateway handles encryption transparently for all traffic crossing between the sites.
- Typical use case analysis: a retail chain has 15 branch stores, each needing constant access to the central inventory database and point-of-sale system at headquarters, with no individual staff member expected to separately start a VPN connection.
  - Every device at each branch needs transparent access, needs a gateway-to-gateway tunnel that's invisible to individual devices, not per-device client software.
  - The connection must be permanent, not something staff start and stop, needs a persistent, always-up tunnel between gateways rather than a user-initiated session.
  - Traffic volume is higher, many staff plus continuous POS and inventory data, needs the sustained throughput dedicated gateway hardware provides.
  - 15 branches means scaling matters, needs a design that's cost-effective per site rather than per individual user.

TYPE 2

# Site-to-Site VPN: Analysis

- Hardware/software: primarily hardware, dedicated routers, firewalls, or VPN concentrators at each site, though software-based gateways are workable for smaller, lower-traffic sites.
- Throughput: must support the aggregate traffic of an entire site, generally a higher and more sustained requirement than a single remote-access user.
- Limitations: requires dedicated hardware and configuration at every site; not suited to individual mobile users, that's exactly what Remote Access exists for instead; a single point of failure unless deliberately designed with redundancy.
- Cost: higher upfront hardware cost per site, but cost-effective per user at scale, historically far cheaper than dedicated leased lines between sites.
- Typical use cases: connecting branch offices to headquarters, or a company's network to a data centre or business partner's network.

# Site-to-Site VPN: Summary

| Dimension | Site-to-Site VPN |
|---|---|
| Hardware/Software | Primarily hardware gateways at each site |
| Throughput | High, must support a whole site's aggregate traffic |
| Limitations | Needs dedicated setup per site; not suited to mobile individuals |
| Cost | Higher upfront per site, cost-effective at scale |
| Typical use case | Branch-to-headquarters, network-to-data-centre links |

# Personal VPN: Overview

- A Remote Access variant where an individual subscribes to a commercial VPN service to encrypt their own internet traffic and often mask their apparent location, not to reach any specific employer's internal resources.
- Typical use case analysis: a journalist frequently works from cafes and airports on public WiFi, needs to protect her research traffic from casual eavesdroppers there, and occasionally wants to appear to browse from a different country, with no need to reach any specific organisation's internal servers.
  - Protection is needed on untrusted public WiFi, not access to a specific company network, needs a VPN whose whole purpose is generic traffic protection to a commercial provider, not organisational resource access.
  - She wants to appear to browse from elsewhere, needs a provider offering exit servers in multiple countries, a feature specific to the personal/commercial model.
  - No corporate IT department is involved, just an individual paying for a service, needs a self-service consumer subscription rather than enterprise gateway administration.

# Personal VPN: Analysis

- Hardware/software and cost: software client apps from a commercial provider, typically a modest consumer subscription, sometimes free with corresponding privacy trade-offs.
- Throughput and limitations: depends heavily on the provider's own server capacity and the user's connection; the bigger limitation is trust, the provider itself can see unencrypted traffic once it leaves their servers toward the real destination, unless that destination separately uses HTTPS.
- Typical use cases: privacy-conscious individuals, protecting traffic on public or untrusted WiFi, bypassing geographic content restrictions.

# Personal VPN: Summary

| Dimension | Personal VPN |
|---|---|
| Hardware/Software | Commercial provider's software client |
| Throughput | Variable, depends on provider server load |
| Limitations | Requires trusting the provider itself with your traffic |
| Cost | Modest consumer subscription, or free with trade-offs |
| Typical use case | Personal privacy, public WiFi, geo-restriction bypass |

# Mobile VPN: Overview

- A Remote Access variant specifically designed to keep a persistent, stable connection as a device moves between networks, WiFi to cellular data, or between cell towers, without dropping the session or forcing re-authentication every time.
- Typical use case analysis: a hospital's home-visit nursing team uses tablets that must maintain a continuous secure connection to patient record systems while staff move between a patient's home WiFi, a vehicle's cellular hotspot, and the hospital's own WiFi.
  - The device moves between fundamentally different network types during one visit, needs a VPN specifically engineered to survive underlying network changes without dropping the session.
  - Interruptions during patient care are unacceptable, needs a persistent virtual identity so the tunnel doesn't depend on a fixed network path, avoiding forced re-logins mid-visit.
  - Access to sensitive patient records demands genuine security, not just convenience, needs the same encryption and authentication as standard remote access, with persistence added on top rather than in place of it.

# Mobile VPN: Analysis

- Hardware/software: software client on the mobile device; the gateway side needs specific mobile VPN support, not just any generic concentrator.
- Throughput and limitations: variable and less predictable than a fixed connection, since it depends on whatever underlying mobile network is currently active; also more complex to implement, and maintaining a persistent connection costs battery life.
- Cost: broadly similar to standard remote access licensing, though mobile-VPN-capable gateway support can add cost.
- Typical use cases: field workers, delivery and logistics personnel, healthcare staff moving through a facility with a tablet, anyone needing an uninterrupted session while genuinely on the move.

# Mobile VPN: Summary

| Dimension | Mobile VPN |
|---|---|
| Hardware/Software | Software client; gateway needs specific mobile VPN support |
| Throughput | Variable, tied to the current mobile network |
| Limitations | More complex to implement; battery cost on the device |
| Cost | Similar to Remote Access, plus possible gateway support cost |
| Typical use case | Field workers and mobile staff needing an uninterrupted session |

# Cloud VPN: Overview

- A Site-to-Site variant connecting an organisation's on-premises network to resources hosted in a public cloud provider, so on-premises systems can reach cloud-hosted resources as though on the same network.
- The cloud provider offers a managed VPN gateway service, such as AWS Site-to-Site VPN or Azure VPN Gateway, that the organisation's own on-premises gateway connects to, typically over an IPsec-based tunnel.
- Typical use case analysis: a manufacturing company keeps its factory floor sensors and control systems on-premises but has moved its data analytics and reporting platform to a public cloud provider, needing continuous, secure data flow between the two without exposing either environment directly to the internet.
  - Two distinct environments need to function as one logical network, needs exactly a site-to-site connection, just with one site being a cloud provider's environment.
  - Data must flow continuously and automatically, needs a persistent, gateway-managed tunnel rather than a manually-initiated user session.
  - The company doesn't want to manage physical VPN hardware inside a data centre it doesn't own, needs the cloud-side gateway to be a managed service the provider operates.

# Cloud VPN: Analysis

- Hardware/software: hardware on the on-premises side; a managed, software-defined virtual gateway on the cloud side, with no physical hardware for the customer to manage there.
- Throughput, limitations, and cost: throughput generally scales well with the provider's pricing tier, but costs typically combine a fixed gateway charge with data transfer fees that grow with volume, and the organisation is tied to that specific provider's gateway configuration and quirks.
- Typical use cases: hybrid cloud architectures, extending an on-premises data centre into the cloud, and disaster recovery replication.

# Cloud VPN: Summary

| Dimension | Cloud VPN |
| --- | --- |
| Hardware/Software | On-premises hardware; managed virtual gateway in the cloud |
| Throughput | Scales with the provider's pricing tier |
| Limitations | Tied to the specific cloud provider's gateway and quirks |
| Cost | Fixed gateway charge plus data transfer fees |
| Typical use case | Hybrid cloud, on-premises-to-cloud extension, disaster recovery |

SUMMARY

# All Five Types, Side by Side

| Type | Connects | Deployment | Throughput | Client Software? | Typical Cost |
|---|---|---|---|---|---|
| Remote Access | One user to a network | Mostly software | Modest per user | Yes | Low per user |
| Site-to-Site | One network to another | Mostly hardware | High, whole-site aggregate | No | High per site, efficient at scale |
| Personal | One user to a VPN provider | Software | Variable, provider-dependent | Yes | Low, consumer subscription |
| Mobile | One roaming user to a network | Software | Variable, network-dependent | Yes | Similar to Remote Access |
| Cloud | On-premises network to the cloud | Hardware on-premises, managed in cloud | Scales with provider tier | No | Gateway fee plus data transfer |

# Summary

- The internet was built for reachability, not confidentiality. A VPN adds the confidentiality, integrity, and authentication the underlying network never guaranteed.
- A tunnel is only encapsulation. A VPN is a tunnel plus security. Not every tunnel is a VPN, but every VPN relies on some form of tunneling.
- VPNs are classified along two independent axes: deployment form (software or hardware) and connection purpose (Remote Access or Site-to-Site, with Personal, Mobile, and Cloud as variants of those two).
- Each of the five types trades cost, throughput, and flexibility differently, and choosing between them is a genuine LO4 evaluation question, not just a naming exercise.

# Coming Up

- Deck B, next: IPsec and SSL VPN in full technical detail, including SSL VPN's modular building blocks, then an overview of how VPNs and tunnels are actually configured on Cisco devices.
- Tutorial worksheet: study-notes-style questions covering today's foundations and all five VPN types. Use it to prep for the mid-semester test.

# References


- Santos & Stuppi, CCNA Security 210-260 Official Cert Guide, 1st Ed., Cisco Press.
- Kaufman, Perlman, Speciner & Perlner, Network Security: Private Communication in a Public World, 3rd Ed., Prentice Hall.
- Cisco Networking Academy, CCNA Security curriculum (VPN technologies chapters).